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STATE-OF-THE-ART REVIEW

Hunger-driven adaptive prioritization of behavior

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In order to survive, an animal must adapt its behavioral priorities to accommodate changing internal and external conditions. Hunger, a universally recognized interoceptive signal, promotes food intake though increasingly well-understood neural circuits. Less understood, is how hunger is integrated into the neural computations that guide nonfeeding behaviors. Within the brain, agouti-related peptide neurons in the arcuate nucleus of the hypothalamus have been found to powerfully stimulate feeding in addition to mediating other hunger-driven behavioral phenotypes. In this review, we compile the behavioral plasticity downstream of hunger and present identified or potential molecular and neural circuit mechanisms. We catalogue hunger's ability to increase exploration, decrease anxiety, and alter social behavior, among other phenotypes. Finally, we suggest paths forward for understanding hunger-driven behavioral adaptation and discuss the benefits of understanding state-dependent modulation of neural circuits controlling behavior.

Hunger biases behavior toward feeding

Animals engage in a wide range of behaviors to fulfill goals critical to life and reproduction. In order to thrive, an animal must appropriately prioritize the behaviors that address their most pertinent needs. The need for food represents one of nature's most fundamental obligations. Hunger, the interoceptive reporter of energy deficit, is a universally recognized signal that triggers a pervasive shift in behavior (Fig. 1). The hungry animal must engage in a significantly altered behavioral complement relative to the fed animal to fulfill this need. Disruption in this complex interaction between energy level and behavior, leading to an abnormal behavioral response to energy deficit or surplus, is a hallmark of eating disorders. However, the mechanisms that underlie this adaptive prioritization of behavior remain largely unknown.

Hunger, while familiar to every animal, can be difficult to operationally define. How do you measure hunger? In humans, you can simply ask them. But what are they assessing internally when they introspect in

Abbreviations
AGRP, agouti-related peptide; AMPK, 5′ adenosine monophosphate-activated protein kinase; Amyg, amygdala; ARC, arcuate nucleus of the hypothalamus; BNST, bed nucleus of the stria terminalis; DBB, diagonal band of Broca; GABA, gamma aminobutyric acid; GPCR, G protein-coupled receptor; Hipp, hippocampus; LH, lateral hypothalamus; MC3R, melanocortin receptor 3; MC4R, melanocortin receptor 4; mPFC, medial prefrontal cortex; NAc, nucleus accumbens; NMDA, *N*-methyl-D-aspartate; NPY, neuropeptide Y; PAG, periaqueductal gray; PAK, P21-activated kinase; POMC, pro-opiomelanocortin; PVH, paraventricular nucleus of the hypothalamus; PVT, paraventricular nucleus of the thalamus; Vgat, vesicular GABA transporter; vHP, ventral hippocampus; Y1R, neuropeptide Y receptor 1; Y2R, neuropeptide Y receptor 2; Y4R, neuropeptide Y receptor 4; Y5R, neuropeptide Y receptor 5.

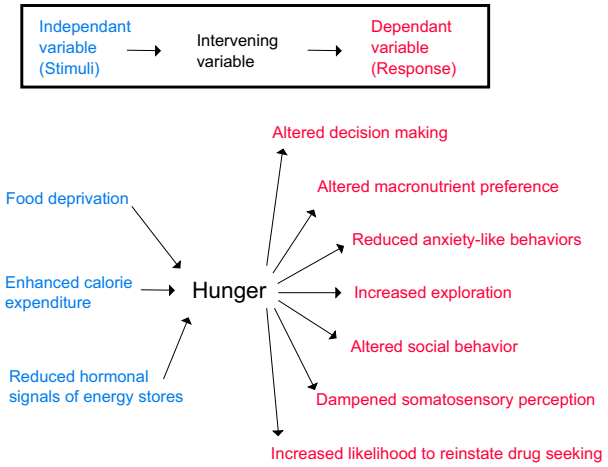

Fig. 1. Hunger as an intervening variable connecting stimuli that alter energy state and the behavioral shifts that support feeding [2,3]. Many stimuli can trigger the need to consume food, and there are similarly many behavioral responses that support the finding and consumption of food. A neural and conceptual organization that allows an intervening variable to collect and communicate these stimuli allows the coordination of a range of behavioral responses to a number of disparate stimuli.

search of an answer? The consequences of hunger can be measured by appetite, or how much food an animal will consume. While difficult to pin down, hunger as a concept serves an important purpose in the study of motivated behavior. From a behavioral perspective, a drive such as hunger can act as an "intervening variable" between a great number of stimuli and responses [1–3] (Fig. 1). For example, a number of stimuli (food deprivation, increased calorie expenditure, hormonal imbalance) can lead to a great number of behavioral responses (food seeking, food consumption, suppression of other behaviors). Assuming a strict one to one stimulus-response organization, these stimulus–response relationships would quickly become cumbersome, both theoretically and biologically. However, an intervening variable between stimulus and response allows a range of stimuli to contribute to the ultimate expression of the appropriate behavioral response. In studying homeostatic behaviors, this intervening variable is most commonly known as a drive. Hunger can be conceptualized as the aggregate neural representation of the many stimuli that can result in increased appetite, food reward, and the behavioral shifts that support feeding.

In this review, we describe behavioral plasticity during energy deficit, focusing on how a hungry animal adapts priorities across behavioral domains beyond feeding. With this resource, we hope that the scientific community can begin to unravel how hunger is integrated into neural circuits guiding behavior, and through this, identify novel therapeutic targets for eating disorders and other maladaptive behaviors. We begin with an overview of the central integrator of energy state, the arcuate nucleus of the hypothalamus (ARC). The arcuate nucleus is linked to many hunger-driven behavioral phenotypes and is a critical communicator of energy state to the brain. We then survey a range of behaviors known to be altered by hunger, including exploratory behavior, anxiety-like behaviors, diet choice, reproductive behaviors, and decision-making. Finally, we suggest a generalized experimental approach to identify novel mechanisms by which hunger spurs adaptive prioritization of behavior.

Arcuate nucleus, a central integrator of energy state

While a drive such as hunger or thirst represent nebulous motivational states that have no single locus, certain brain regions like the ARC play an outsized role in communicating these states to the nervous system. The arcuate nucleus is a region in the mediobasal hypothalamus that contains multiple neuronal subtypes responsible for controlling distinct aspects of feeding and hunger-related behaviors. Two arcuate cell types, cells that express agouti-related peptide (AGRP) and cells that express Pro-opiomelanocortin (POMC), act in opposition to promote and reduce energy intake, respectively. By integrating signals from circulating hormones, ascending circuits from the gut and brainstem, and incoming sensory information, these neurons coordinate neural systems to ensure maintenance of energy homeostasis [3,4]. Multiple experimental approaches have shown AGRP neuron activation invigorates a behavioral program to find and consume food. Both optogenetic [5] excitation and chemogenetic [6] excitation dramatically enhance food intake in fed animals. Conversely, chemogenetic silencing of AGRP neurons is able to rapidly attenuate feeding in hungry mice [6]. Additionally, AGRP neurons prepare the animal for feeding by altering behavioral strategy [7,8] while also coordinating peripheral metabolism [9,10]. Ablation of AGRP neurons in adult mice results in a dramatic reduction in food intake, potentially leading to death [11,12]. However, if AGRP neurons are ablated in neonates, a yet unknown compensatory mechanism results in normal feeding behavior and survival [11]. The necessity of these neurons in adult animals indicates their central role in connecting energy state with an appropriate behavioral response. AGRP neurons have provided immense utility in allowing

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researchers to produce an "artificial hunger" state; however, there are more neuron populations and hormones that contribute to the central representation of energy deficit [3]. Here, we focus on AGRP neurons mainly because they have been most extensively studied using the genetically encoded tools (Cre-dependent opto- and chemogenetics) that have allowed the dissection of a wide range of hunger-driven behavioral phenotypes [13]. As the main focus of this review is hunger-driven adaptations, we will focus on the AGRP neuron component of the arcuate nucleus. The arcuate nucleus as a whole, as well as the concept of satiety and associated neural correlates, has been comprehensively reviewed elsewhere [3,4].

In response to an energy deficit, AGRP neurons engage several synaptic plasticity mechanisms resulting in significantly upregulated action potential firing frequency [14]. Excitatory synaptic input onto AGRP neurons is likely upregulated via multiple mechanisms. Different groups have identified presynaptic enhancements in glutamate vesicle release probability and postsynaptic dendritic spinogenesis. Presynaptic glutamate release probability is enhanced by ghrelin signaling, which engages a 5′ adenosine monophosphate-activated protein kinase (AMPK)-dependent positive feedback loop [15]. This presynaptic enhancement is negatively regulated by leptin, which is able to inhibit presynaptic release in response to increasing energy stores following food intake. Postsynaptically, activation of *N*-methyl-D-aspartate receptors leads to spinogenesis which also acts to enhance excitatory transmission onto AGRP neurons [16]. This spinogenesis is downstream of postsynaptic AMPK-mediated activation of p21-activated kinase (PAK), and activation of AMPK in AGRP neurons is sufficient to drive feeding and weight gain [17]. Ultimately, these changes result in a significantly increased firing rate of AGRP neurons, presumably leading to increased release of neurotransmitters, gamma aminobutyric acid (GABA) and neuropeptides, in downstream regions. However, the significance of these mechanisms in promoting the behavioral phenotypes of AGRP neurons has not been studied.

The production of multiple neurotransmitters by AGRP neurons leads to diverse signaling capabilities. They utilize the inhibitory neurotransmitter GABA, as well as multiple neuropeptides, including AGRP and neuropeptide Y (NPY) (Fig. 2). AGRP produces its behavioral effect by acting as an inverse agonist at melanocortin G protein-coupled receptors (GPCRs) melanocortin receptor 3 and 4 (MC3R and MC4R) [3]. These receptors are activated by the POMC product alpha-melanocyte-stimulating hormone and are typically coupled to Gαs G proteins. Thus, AGRP acts to enhance appetite by reducing endogenous, satiety promoting, MC3R and MC4R signaling. NPY acts through a family of GPCR Y receptors known as Y1, Y2, Y4, and Y5 in humans [18]. All Y receptors are typically coupled to Gαi G proteins and exert an inhibitory influence. Postsynaptic Y1 and Y5 receptors are most commonly associated with the food intake promoting action of central NPY [18]; however, the presynaptically localized Y2 receptor has been found to mediate anxiolytic and antidepressant properties of NPY [19]. Further work will need to examine these signaling systems in greater detail in downstream

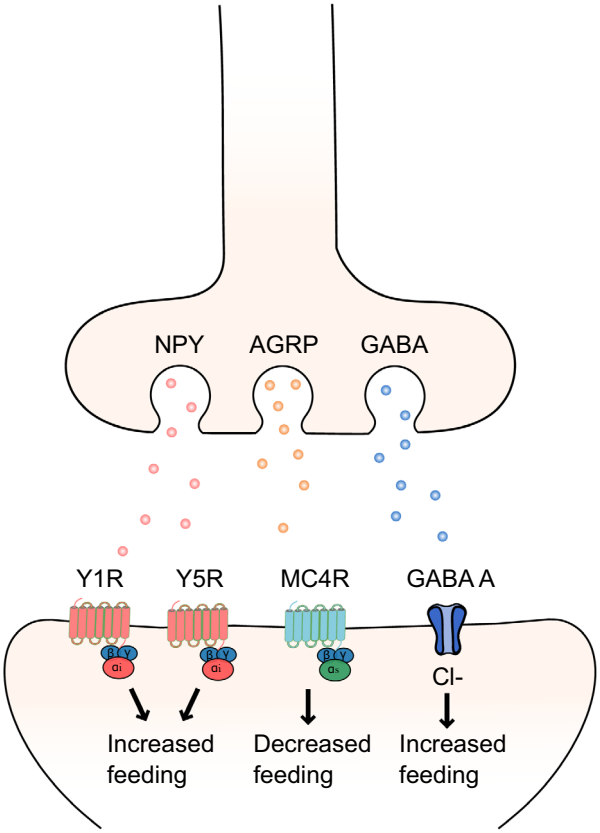

Fig. 2. AGRP neurons release NPY, AGRP, and GABA to communicate low energy state. NPY acts through the Y family GPCRs, of which Y1 and Y5 have been described as the primary effectors of NPY's effect on feeding behavior. AGRP acts as an inverse agonist at melanocortin GPCRs, including MC4R and MC3R. GABA acts at the GABA A receptor, a ligand-gated ion channel, to allow immediate inhibition of postsynaptic targets. AGRP neuron neurotransmission has mainly been studied in the context of feeding behavior, and many AGRP neuron phenotypes independent from feeding are not associated with a specific transmitter or receptor system.

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regions in order to more clearly define the generation of different behavioral phenotypes.

Studies utilizing different genetic manipulations to perturb specific neurotransmitters released by AGRP neurons have identified distinct roles for these signaling systems. Similar to AGRP neuron ablation, constitutive knockout of *Agrp* or *Npy* results in no long-term feeding or body weight phenotype [20,21]. Conversely, ablating GABA transmission from AGRP neurons utilizing an *Agrp-Ires-Cre, Vgat*$^{flox/flox}$ cross results in a reduction in both body weight and ghrelin-stimulated food intake [22]. In light of the underwhelming phenotype of the single knockout strains, a triple knockout mouse was generated lacking *Mc4r* and with *Agrp-ires-Cre* driving AGRP-specific knockout of *Npy* and *Vgat* [23]. Chemogenetic stimulation of AGRP neurons in this triple knockout mouse failed to evoke food intake, indicating the combined necessity of these three signaling systems to allow AGRP neuron stimulation induced feeding behavior [6]. Further double knockout crosses identified GABA and NPY signaling as sufficient to drive the acute increase in food intake characteristic of AGRP neuron stimulation on a timescale of minutes [23]. Indeed, analysis of single *NPY* knockout mice finds that NPY is necessary for AGRP neuron stimulation to promote short-term increases in food intake [9]. AGRP signaling through MC4r on the other hand was found to drive a more chronic increase in food intake on a timescale of hours [23]. While many studies of AGRP neuron modulation of feeding behavior utilize concurrent stimulation paradigms, endogenous AGRP neuron activity seems to be dramatically reduced upon food intake. In paradigms mimicking this activity pattern by utilizing optogenetic prestimulation prior to food introduction, NPY signaling is necessary for enhanced food intake [24]. In addition to these three well-studied neurotransmitters, there appear to be other neuropeptides also produced in AGRP neurons, including the products of the propeptide ProSAAS [25]. Importantly, the existence of GABA, NPY, and AGRP in the same cell type, all contributing uniquely to feeding behavior, illustrates that a single cell type can contain multiple transmitters each participating in unique interactions toward a shared phenotype. As the role of individual neurotransmitters has only been assessed for the food intake phenotype of AGRP neuron activation, it is possible that a variety of neurotransmitters may be key to initiating other behavioral phenotypes downstream of AGRP neurons.

Agouti-related peptide neurons send projections to brain regions both within and outside the hypothalamus, but only some of these terminal fields are able to stimulate food intake (Fig. 3). Within the hypothalamus, AGRP neurons project to the paraventricular nucleus of the hypothalamus (PVH) as well as the lateral hypothalamus (LH) [26]. Beyond the hypothalamus, AGRP terminals reach the bed nucleus of the stria terminalis (BNST), the paraventricular nucleus of the thalamus (PVT), the amygdala (Amyg), and the periaqueductal gray (PAG) [26]. Additionally, a projection from AGRP neurons to the nucleus accumbens (NAc) has been reported [27]. A number of these projections are able to enhance food intake (Amyg, BNST, PVH, PVT, LH) [8,26]. Other projection fields appear to modulate different behaviors, allowing enhanced food seeking and feeding through the inhibition of competing behavioral drives, such as projections to the parabrachial nucleus (PBN) which mediate the suppression of pain, projections to the Amyg that mediate territorial aggression, and projections to the BNST that mediate aspects of glucose homeostasis [8,10,28]. Additionally, AGRP neurons have been found to be able to trigger cascades of circuit changes leading to phenotypes in regions several synapses away, such as biasing food cue responses in the insular cortex via a circuit containing the PVT and Amyg [29]. The wide reach of these hunger neurons suggests a critical role of energy state in neural computations made across the brain. Through AGRP neuron modulation of downstream circuits, animals are able to consider their current hunger level when deciding to engage, or avoid, specific behaviors.

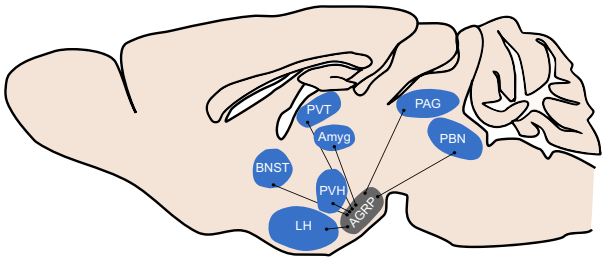

Fig. 3. First-order projection targets of AGRP neurons in the arcuate nucleus. AGRP neurons exhibit a broad range of projection targets, contacting regions involved in a diverse set of behaviors. Interestingly, there appear to be different neuronal populations of AGRP neurons for each projection target. These projection targets support AGRP neuron modulation of a wide range of behavioral phenotypes, including influencing food consumption through terminal fields in the PVH, LH, BNST, and PVT. Terminal fields in the Amyg and PBN have been found to influence territorial behavior and pain-associated behaviors, respectively. Mesolimbic regions such as the ventral tegmental area are thought to play a role in food reward and food-seeking behavior, but exact mechanisms downstream of AGRP neurons are unknown.

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In addition to aforementioned regulation by long-term energy state, AGRP neurons are also regulated on distinct timescales by both external sensory information and internal information communicating nutrient intake in the gut. Upon food presentation, AGRP neurons exhibit rapidly reduced intracellular calcium as well as reduced action potential firing frequency [30–32]. This immediate response, before food is ingested, is thought to rely on sensory information transmitted to AGRP neurons and has been found to be transient in nature. A similar, slower effect can be produced by intragastric nutrient infusion or gastric distension [33–35]. Further, administration of satiety hormones such as CCK and PYY is able to reduce AGRP neuron calcium response [33,34]. Finally, leptin is able to durably inhibit AGRP neuron calcium on a timescale of hours [33]. Together, these results suggest that when a hungry animal encounters food, AGRP neurons are suppressed in distinct temporal phases by differing mechanisms that exhibit individualized staying power. Interestingly, these mechanisms seem to be disrupted following exposure to a high-fat diet, suggesting a mechanism by which calorically rich foods produce long-term changes in the neural response to nutrient detection and intake [36,37]. These results indicate that we still have much to learn about how current and future whole-body energy state is assessed on a second by second basis, and how that communication is altered in disease states.

Hunger-driven behavioral plasticity

Intuitively, we can appreciate that hunger has a powerful ability to promote consumption of food. The ability of hunger to enhance food intake by decreasing satiety and increasing food reward is a well-studied phenomena [4,38,39]. Less obvious is the impact hunger has on a wide range of behaviors beyond simply increasing food consumption. As an animal's energy-deficit increases, finding food becomes more urgent than competing needs. From anxiety-driven behaviors to reproductive behaviors, the behavioral complement of a hungry animal drastically shifts in order to prioritize finding and consuming food (Fig. 1) [40]. Despite the intuitive nature of this assertion, very little is known about how hunger modulates the neural circuits controlling competing behaviors in order to adapt to new priorities. In this section, we seek to outline the current behaviors known to be altered by hunger and discuss any implicated mechanisms for this behavioral plasticity.

Hunger encourages foraging and exploration of the environment

In order to survive in the wild, animals must engage in foraging behavior that allows them to access food without exposing them to unnecessary danger. The foraging behavior of different wild animals has been a subject of intense research in the fields of behavioral and evolutionary ecology. The evolutionary fitness of an animal depends heavily on how well it is able to find food. Selective pressure has resulted in food-seeking behavior that is tightly regulated in order to ensure close to optimal strategy, meaning maximizing net energy gain. This concept has been modeled extensively as optimal foraging theory, and animal behavior during a bout of foraging has been found to hew closely to these models [41,42]. In fact, the behavior of a foraging animal has been described using economic models that minimize resources spent in the acquisition of further resources. This is described as the marginal value theorem [43]. However, to use this model to describe the long-term behavior of an animal, a critical variable to include in these models is the state of the animal's energy reserves, as well as the risk of predation in foraging regions [44]. Clearly, a food-deprived animal will make different foraging decisions than a relatively replete conspecific.

The influence of hunger on an animal's locomotor and exploratory behavior, which underlie foraging, is highly dependent on the environmental context in which the behavior is assessed. Overarching behavioral patterns differ depending on whether the environment is novel, whether food is currently present, and the presence of other novel nonfood objects. As such, different experimental preparations will produce different results when examining this broad phenotype. Of principal importance is the animal's experience with food in the experimental context. When food is not present, if the animal has found food in the area before, hunger is likely to enhance locomotor behavior. This is seen in home cage monitoring, where food deprivation enhances locomotor behavior [45]. However, when food is present, locomotor activity itself is not enhanced, but is instead directed toward areas with food and away from those without [6,46]. Interestingly, when presented with food, an animals' foraging behavior does not immediately cease, despite food already being available [45]. Similarly, while it does decrease, investigation of a nonfood object does not cease in the presence of food, even in a hungry mouse. This indicates separate behavioral responses for searching for food and consuming available food. These results seem to be sensitive to mouse strain, as different strains

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exhibit differential changes in locomotor activity, as well as body temperature regulation, in response to food deprivation [47]. However, broadly the characteristic impact of hunger over exploration behaviors is conserved, even across species. In zebrafish, hunger encourages chase of smaller objects, while larger, potential predator objects become less aversive [48]. This has been traced mechanistically to energy state-dependent changes in visual processing in the fish tectum [48]. Similarly, the nematode *Caenorhabditis elegans* is more willing to cross an aversive hyperosmotic barrier when food deprived [49]. Clearly, the motivational impact of hunger to encourage exploration is a highly conserved phenotype. However, because of the dependence of this increased exploration on context, we consider this behavioral shift to be likely due to a number of different changes in multiple circuits, in likely both sensory and behavioral circuits, as opposed to a simple upregulation of locomotor circuits.

Hunger alters decision-making to promote generalized acquisition and decrease risk aversion in economic choice

Hunger has been found to alter decision-making, increasing impulsivity and biasing behavior toward acquisition of both food and nonfood objects. Following meal ingestion, the relative change in plasma ghrelin levels predicts the change in economic risk aversion in human participants [50]. Hunger also has a significant impact on delay discounting, or the ability to restrain impulse and refuse an immediate reward for a greater reward later. Hungry participants show greater devaluation of future rewards compared with immediate ones. Interestingly, this devaluation occurs regardless of if the reward is a food or nonfood object, indicating a change in global reinforcement [51]. This effect on decision-making is not limited to economic decisions, but also applies to social decision-making. Manipulating nutritional content to reduce protein intake of meals can result in human participants increasing their propensity to hand out social punishment for norm violations [52]. Additionally, low blood glucose levels are associated with greater reported support for social welfare, but also with reduced rates of direct sharing of resources [53]. These changes in social decision-making describe a hungry individual who encourages prosocial behavior and adherence to norms for the group, but in individual decisions favors retaining resources for themselves. Together these results characterize an interesting behavioral phenotype of hunger, one that encourages acquisition and retention of resources, via individual and social means, at the cost of maintaining reduced risk exposure. These broad phenotypes have significant implications for both feeding and nonfeeding decisions made while hungry, and also suggest that prioritizing long-term well-being becomes more difficult when hungry. Identifying the mechanisms underlying these phenotypes would generate therapeutic targets that could enhance treatment compliance when attempting to lose weight but could also reduce potential "off-target" effects of dieting on nonfeeding decision-making.

Risk aversion and impulsivity are broad constructs rather than discrete behaviors, and thus are not mediated by a single neural circuit. Instead, there are a variety of neural systems that can contribute to these phenotypes, and thus likely a range of mechanisms and circuits underlying hunger's effect on decision-making. One neural locus able to impact this behavioral phenotype is the medial prefrontal cortex (mPFC). A large body of work has described the mPFC as critical for constraining impulsivity [54], and mPFC neurons are able to drive food consumption and food-seeking behavior [55–57], indicating that these neurons may be sensitive to hunger in order to assist in their control of feeding behavior. Notably, both patients with anorexia nervosa and bulimia nervosa exhibit altered mPFC activity in response to food images [58]. Patients with anorexia in particular have been described to engage in more ridged decision-making that correlates with altered neural activity in prefrontal regions [59]. Additionally, patients with obesity exhibit decreased functional connectivity between the mPFC and associated executive control regions [60]. It is possible that hunger modulates neural activity within the mPFC in order to modulate decision-making and impulse control and that these mechanisms are dysregulated in disease states.

Recent work has identified a potential molecular and circuit mechanism for enhanced impulsivity during food deprivation. The ventral hippocampus (vHP) receives projections from LH melanin-concentrating hormone neurons and has been found to enhance impulsive responding for food reward in food-deprived animals [61]. It is still unclear whether this regional signaling system is able to mediate a generalized increase in impulsivity beyond food seeking. However, it is likely that neural computations within the vHP are modulated by energy deficit. Indeed, mPFC inputs from the vHP are modulated by glucagon-like peptide 1 (GLP-1) [62], a powerful satiety hormone [63], indicating some degree of control by energy state. Together, these results suggest that an LH to vHP to mPFC circuit is likely responsive to hunger, and

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hunger-dependent modulation of this circuit could contribute to changes in impulsivity.

Another region of interest potentially responsible for hunger-driven changes in impulse control is the NAc. The NAc receives a significant projection from the mPFC and has been found to be important in constraining impulsive action. Lesion studies of the NAc have found that the NAc core in particular is critical to restraining impulsive action [64,65]. Parvalbumin-expressing interneurons within the NAc exhibit sustained activity when an animal is constraining an impulsive response [66]. Perhaps these neurons, or inputs onto these neurons, are altered by hunger in order to reduce the ability to constrain impulsive responding.

We predict that in order to understand the etiology of eating disorders, which are characterized by alterations in decision-making, we must understand how hunger impacts circuits controlling decision-making. Interestingly, the brain regions described above as each contributing to impulsivity and decision-making exist within an interconnected network (Fig. 4). These regions receive a variety of modulation from energy state-dependent neurotransmitters, which likely directs the interaction of these regions to impact impulsive phenotypes. We believe that this hunger-driven modulation is disrupted in these disorders, leading to inappropriate behavioral responses to energy deficit and surplus. Broadening our understanding of how hunger modulates these decision-making circuits could identify novel biomarkers or therapeutic targets to treat eating disorders.

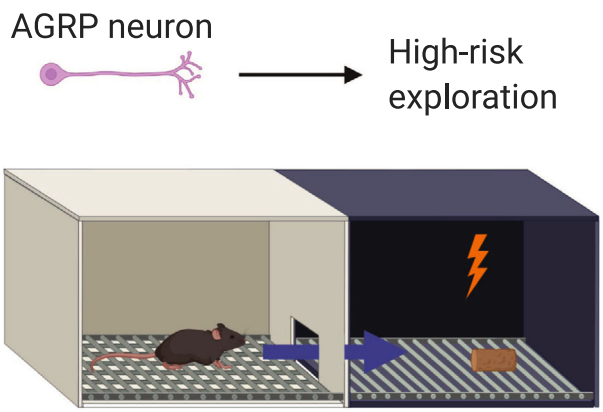

Fig. 4. Artificial activation of AGRP neurons is able to spur animals to enter dangerous areas to forage for food. This mimics the food-deprived phenotype, regardless of whether the food is actually available or out of reach of the animals. This experiment describes a general phenotype of hunger, willingness to acquire resources is enhanced, leading to an increase in risky behavior. Created with BioRender (Toronto, Canada).

Hungry animals display reduced anxiety-like behaviors

The perception of threat is a necessarily salient motivational stimuli, allowing animals to avoid potential danger. When facing starvation, animals must reassess what risks they are willing to take on in order to survive. Basal anxiety state, assessed by elevated plus maze and open-field center time, has been shown to be decreased in food-deprived animals [67], an effect that has also been produced with AGRP neuron stimulation [45]. Thus, a hungry animal exhibits reduced anxiety-like behaviors, displaying a willingness to enter into threatening environments in the face of a potentially more dangerous internal threat. This increased willingness to enter into potentially dangerous situations is likely an effort to increase the probability of finding food. In food-deprived mice trained to associate an area with a foot shock, pairing the foot shock context during training with either accessible or inaccessible food will result in more time spent in the shock-associated area (Fig. 5) [8]. Conversely, fed mice will avoid the shock-associated area. Similarly, in a resident-intruder test, a food-deprived animal will display reduced aggressive behavior upon the addition of an intruder mouse [8]. This reduced aggression could be interpreted as a decreased perception of intruder threat or a decreased desire to defend territory that is considered less valuable because it no longer provides food. Together, these energy state-dependent behavioral adaptations enhance the probability of finding food while reducing energy expended on other competing drives.

Both reduced anxiety-like behavior and reduced territoriality have been linked to activity in AGRP neurons, and the effect of hunger on territorial aggression has specifically been described to be mediated by a circuit connection between AGRP neurons and medial Amyg neuropeptide Y receptor 1 (Y1R) expressing cells [8,68]. Together, these results demonstrate that hungry animals are willing to enhance their exploration into potentially dangerous environments as well as reduce the amount of energy they spend defending a territory. It would be interesting to explore how quickly the territoriality behaviors return upon re-exposure to food, considering the longer scale time course of NPY-mediated inhibition in other systems. It seems logical to expect territoriality to increase in order to defend the scarce resource, but such a timescale would imply a faster switch in circuit function

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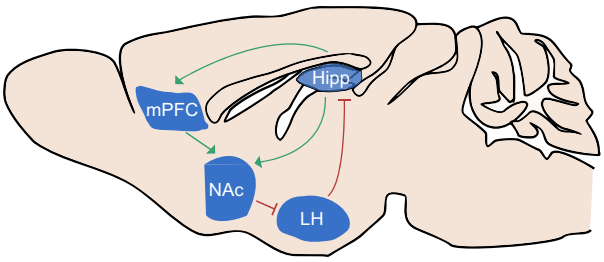

Fig. 5. Interconnected network of brain regions involved in decision-making and impulsivity. Each of these regions has been described as contributing to impulsivity in decision-making. The mPFC is critical to decision-making, and activity in this region is altered in patients with eating disorders. The mPFC receives input from the vHP (Hipp), an area in which modulatory input from the LH has been found to influence impulsivity in the delayed responding task in rodents. Both the mPFC and the Hipp project to the NAc, a region in which interneurons have been found to constrain impulsive actions. The NAc sends a projection to the LH, potentially influencing output to the Hipp. Thus, we propose that neural communication through this network is shaped by the influence of whole-body energy state to ultimately impact decision-making.

than a typically slower decay of neuropeptide mediated inhibition. Beyond territoriality behavior, the specific circuits and molecules underlying the varied anxiolytic effects of hunger are unclear. Recent work indicates that an AGRP neuron projection to LH orexin neurons is able to suppress avoidance of aversive environments, but this has not been described in an energy state-dependent manner [69]. The LH also sends a projection to the diagonal band of Broca (DBB), making monosynaptic inhibitory connections onto cells that encode aversive environmental cues [70]. Whether these DBB projecting LH cells receive direct AGRP input, or are modulated by energy state via an indirect method, is unknown. Another region potentially important to hunger-driven suppression of anxiety-like behavior is the PVT. Silencing posterior PVT neurons alters the balance between food-seeking behavior and threat avoidance behavior [71], suggesting a role for the PVT in balancing conflicting behaviors and mediating adaptive behavior. Although AGRP neurons project to the PVT, it is unclear whether PVT neurons mediating this adaptive behavioral response are modulated by input from the arcuate nucleus. Finally, AGRP neurons also send a projection to the BNST, an area integral to anxiety and negative affective states. However, any behavioral role of this projection field in anxiety-like behaviors is unknown. It is clear that hunger and the expression of anxiety-like behaviors are deeply intertwined, and there are likely numerous mechanisms in multiple brain regions that contribute to the balanced expression of these behaviors.

Further work describing how hunger is able to dampen anxiety could lead to novel therapeutic strategies to combat maladaptive levels of stress. The endogenous suppressors of anxiety produced by hunger may prove clinically useful, and they may be disrupted in individuals prone to abnormal levels of stress. Importantly, disorders of anxiety are highly comorbid with eating disorders [72]. Could these maladaptive eating habits result in part from an imbalance in the hunger-driven modulation of anxiety circuits? Enhancing our understanding of how hunger intersects with anxiety is likely to result in translational gains for both eating and anxiety disorders.

Hunger dampens response to negative somatosensory stimuli

While hunger warns an animal of impending harm, pain can signal an immediate or ongoing insult that can require attention to avoid continued damage. Acute pain serves to alert the animal of a significant immediate threat, while inflammatory pain signals an ongoing, but potentially less immediate, threat. Hunger can suppress both the affective and behavioral responses to inflammatory pain, allowing the animal to focus on obtaining calories necessary for survival [28]. This suppression of inflammatory pain relies on NPY Y1R signaling within the PBN. The endogenous source of NPY for this signaling is likely AGRP neurons, as AGRP neuron stimulation is able to replicate this phenotype. Similarly, AGRP neuron stimulation is able to suppress scratching behavior in response to pruritogenic stimuli [73]. This suppression is also present in food-deprived animals. Together, these results indicate that AGRP neurons, stimulated by energy deficit, can act to dampen somatosensory stimuli that would otherwise distract the animal from finding and consuming food. While hunger-induced suppression of the behavioral response to inflammatory pain has been localized to the PBN, the neural locus for the suppression of scratching behavior is unknown. Further characterization of this phenomena in different metabolic states could identify disrupted regulation of somatosensory stimuli and could identify novel therapeutics for somatosensory disorders such as chronic pain.

Hunger alters diet choice and taste preference

Visionary work by Curt Richter in the 1940s extended the definition of biological homeostasis beyond the

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internal milieu and into the realm of animal behavior. Richter induced salt appetite in rats via adrenalectomy, a procedure that causes the kidneys to excrete large amounts of salt, resulting in death in 5–18 days. However, if rats have been acclimated to a sodium chloride laced water source, they would compensate for the salt loss with enhanced intake of the sodium chloride solution [1]. Animals would select the salt-enriched solution up to a concentration of one part salt to 33 000 parts water, indicating the strength of this appetite to drive behavior. Further, this behavioral compensation for salt appetite could be eliminated by severing the cranial nerves that communicate taste [1]. These results demonstrated that the internal needs of the body powerfully alter behavior in order to maintain homeostasis. They also set the stage for future work examining the role of hunger in changing diet preference.

Over the course of a day, animals change their dietary preferences in order to prepare for the relevant energy demands. While over a long period of time, intake of fat, protein, and carbohydrate are relatively even when animals are allowed to choose from diets enriched in each macronutrient, upon examination of shorter timescales, temporal patterns of macronutrient consumption emerge. At the onset of the active phase, rats preferentially increase their intake of carbohydrates, a preference that eventually subsides [74]. This relative preferences may indicate different values for macronutrients at different phases in the active cycle. In the early portion of the active phase, diet choice may bias toward the more immediate goal of fulfilling short-term energy requirements. By the late portion of the active phase, diet choice may be directed toward macronutrients that generate more long-term satiety and serve more distant energy needs in the upcoming inactive phase. This macronutrient preference can be shifted by altering available stored energy. If food deprivation is broken just prior to the onset of the active phase, animals exhibit a strong increase in preference for food containing high amounts of fat [74–76]. If this food deprivation is instead broken in the late dark period, there is a much smaller increase in overall food intake and no clear preference in macronutrient intake [74]. Together, these results indicate that hunger can have differential effects on macronutrient preference depending on the current phase of the active cycle. This suggests a circadian mechanism interacting with the mechanisms controlling macronutrient preference.

Hunger has also been found to alter taste perception. Specific tastes signal nutrient composition of food, such as sweet tastes indicating sugar and umami tastes indicating protein. Conversely, bitter and sour tastes communicate unsavory aspects of food. Together, these tastes communicate internal values of food that are able to positively or negatively influence consumption. In energy deficit, the perception of these tastes is altered [77,78], likely to encourage consumption of food. This difference in taste perception is regulated by AGRP neuron projections to the LH, with different LH outputs regulating the perception of appetitive and aversive tastes [79]. It is likely this hunger-driven alteration in taste perception is important to food consumption, and contributes to nutrient preference in hungry animals. It is unknown whether this modulation is imbalanced in the context of obesity, where the enhancement of tastes indicating high energy content could be abnormally elevated, resulting in overconsumption.

Hunger dampens social and reproductive behaviors

Finding a mate and reproducing requires significant energetic cost and thus is highly sensitive to energy stores and food availability. In times when energy is scarce, suppressing reproductive drive allows the individual to conserve energy in addition to ensuring offspring do not starve. To achieve this, hunger seems to specifically dampen the initiation of reproduction. First, it reduces approach to the mating partner. Second, it lowers the probability of initiation of sexual behavior. In tests examining the behavior of animals around reproductively relevant cues, hungry animals spend significantly less time around these cues compared with their fed counterparts [80]. When receptive partners are present, sexual behaviors are reduced. Food-deprived females display longer latencies to lordosis in addition to shorter durations [81]. They are also much less likely to mate than freely fed and re-fed females [82]. Food-deprived males are also slower to initiate mating [80] and exhibit decreased copulatory rate [83,84]. This hunger-driven reduction in mating behaviors is likely downstream of AGRP neurons, as optogenetic activation of AGRP neurons produces a very similar phenotype to food deprivation [68]. Food deprivation also impacts parenting behavior. Food-deprived females exhibit decreased maternal behaviors, including reduced nest building and pup retrieval [85]. This phenotype is mediated by an AGRP neuron to medial preoptic area Vgat+ circuit. Together, these results describe a generalized inhibitory effect of hunger on mating and parenting behaviors.

While the overall direction of this effect is clear in both sexes, we still have an incomplete understanding

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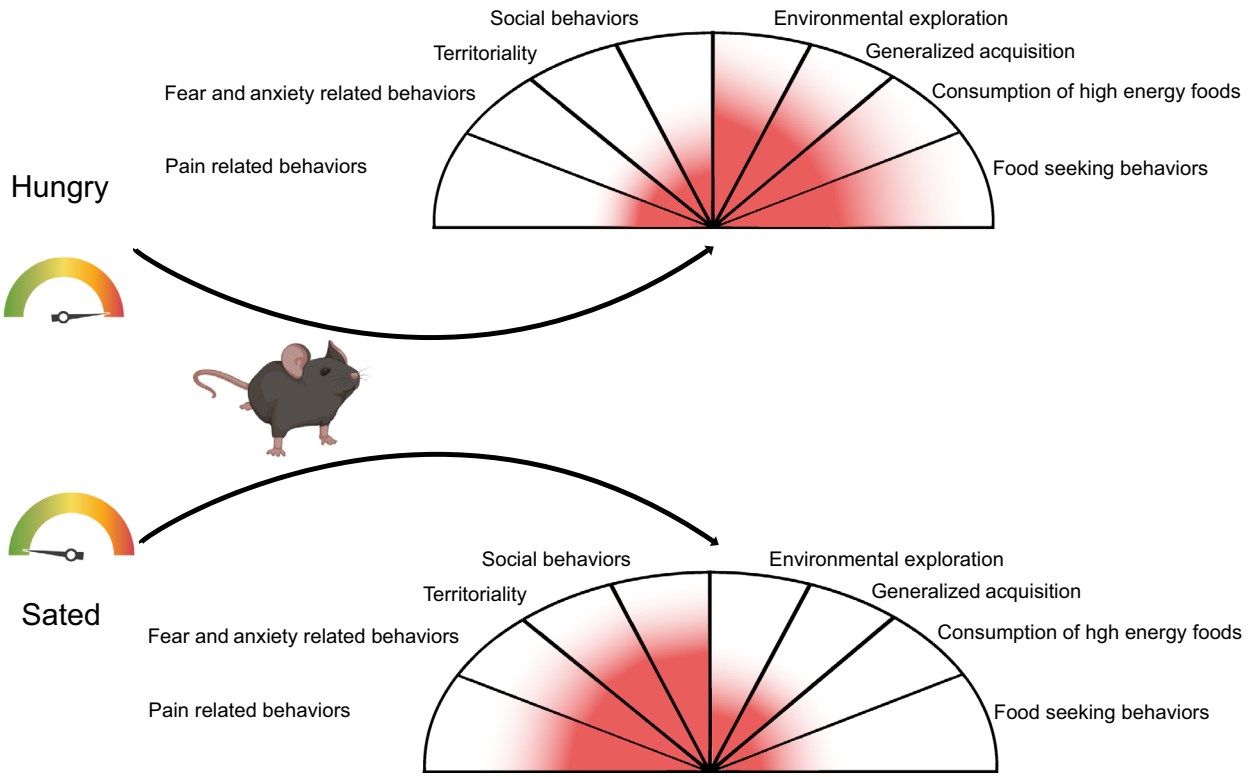

Fig. 6. Signals communicating whole-body energy state shift activity in central circuitry resulting in adaptive prioritization of behavior depending on the most pertinent needs. Certain behaviors are suppressed while others are enhanced, increasing the probability of finding food and reducing energy expended on other tasks. Underlying this behavioral plasticity is likely extensive modulation of a wide range of neural circuits, only some of which has been uncovered thus far. Made with BioRender (Toronto, Canada).

of what neural circuits in either sex are modulated by hunger to produce this behavioral suppression. Further, it is unknown whether these mechanisms are disrupted in eating disorders, and how this impacts sexual behavior. Male sexual behavior is sensitive to NPY Y1R signaling in the dorsal raphe nucleus, and antagonism of this signaling is able to partially restore sexual behavior in food-deprived males [84]. Additionally, one potential circuit mechanism for hunger's influence over female sexual behavior involves AGRP neuron projections to kisspeptin neurons. This circuit has the ability to suppress female fertility, but it is unknown whether this mechanism is engaged in periods of energy deficit [86]. AGRP neurons also send projections to the PVH, a region that contains neurons expressing oxytocin, a neuromodulator that has been found to be critical for sexual and maternal behaviors [87,88]. But this projection has not been linked to appetite-dependent regulation of sexual or maternal behaviors. Additionally, while the above mechanisms represent modulation at the level of circuits controlling behavioral expression, it is also possible that sensory circuits governing sensitivity to social and reproductive cues are also modulated by hunger.

Recent work has sought to address the question of how hunger impacts different social behaviors in an ethologically relevant context, as has been elegantly reviewed [40]. Outside of the laboratory, animals encounter these competing behavioral choices concurrently, meaning the pursuit of one behavior typically occurs at the expense of the pursuit of another. This fact is not well represented in many laboratory behavior experiments, where a deliberately reductive environment isolates the performance of a specific behavior. When the role of hunger is examined in the context of competing motivational drives, a highly complex picture emerges [68]. When testing the ability of a resident-intruder paradigm to stimulate social interaction, hunger was found to significantly decrease social interaction with either a male or female intruder [68]. However, intrusion was able suppress feeding behavior in a resident mouse, when the intruder was female. Feeding on the part of the intruding mouse was found to be strongly suppressed when entering into a territorialized cage, even when the resident was not present. Generally,

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these findings follow theme that hunger encourages the prioritization of feeding over other motivations, but they raise the importance of studying these behaviors in ethologically relevant contexts.

Hunger triggers reinstatement of drug seeking

Drug addiction is a persistent disorder characterized by repeated periods of abstinence and relapse [89]. This has led to the development of experimental methodology in which animals learn a drug reward association or seeking behavior, undergo extinction, and then have that behavior reinstated by a challenge, typically either drug or stress. Chronic food restriction and acute food deprivation have both been found to reinstate drug reward associations and drug-seeking behavior [90,91]. Interestingly, chronic food restriction appears to induce relapse via different mechanisms than acute food deprivation [92]. Reinstatement of heroin seeking by acute food deprivation appears to rely on corticotropin-releasing factor signaling [93], as well as dopamine D1 receptor signaling [94,95]. Acute food deprivation is known to elevate plasma corticosterone [96], a hormone known to mediate stress-induced reinstatement in other paradigms [93]. Interestingly, reinstatement of cocaine seeking via acute food deprivation requires corticosterone [97], while reinstatement of heroin seeking by acute food deprivation does not [93]. This suggests that the stress induced by acute food deprivation engages multiple distinct pathways to alter animal behavior. Additionally, it is possible to augment ongoing heroin self-administration and reinstate heroin seeking by acute administration of NPY [98]. This may be emblematic of a more general distinction between heroin- and cocaine-seeking behaviors, where heroin seeking is more susceptible to regulation by energy state-dependent transmitters. This is potentially due to the fact that endogenous opioids regulate feeding behavior [99], creating shared neural substrates governing these behaviors. While the ability of different stress-related signals in promoting food deprivation-induced relapse is clear, it is unknown what circuits are necessary to mediate this effect. Further, it is unknown whether more classical signals of energy state are sufficient to induce this relapse, or if the stress component of food deprivation alone is the singular cause.

Understanding hunger as a critical player in nonfeeding behavioral plasticity

In approaching these questions, we suggest a generalized experimental strategy. First, there should be an initial behavioral characterization of a phenotype presenting a difference between food-deprived and fed animals. This could be followed by a screen or hypothesis-driven search for a central circuit element or transmitter system responsible for this phenotype. This could be achieved via a number of techniques, including proteomic screen to examine brain region-specific concentrations of neuropeptides, transcriptomic screen to examine cell type-specific expression of transmitter mRNA, or cFOS (or other immediate early gene) screen to examine cell type-specific markers of neuronal activity. These screening methods have the potential to greatly enhance our understanding of hunger-driven changes in neural dynamics. While this review has focused primarily on AGRP neuron phenotypes, there are likely other neural populations that exhibit changes in activity or transmission. Next, it is important to test whether this transmitter or circuit element is necessary or sufficient to produce this phenotype. Finally, mechanistic description of how this circuit element or transmitter impacts its environment, whether that be a neural circuit or a cell. While significant advances have been made in understanding the endogenous activity of AGRP neurons during feeding behavior, it would be interesting to examine these neurons in other contexts. Further, we still have little understanding of how this activity at the soma translates into downstream signaling systems. Utilizing *in vivo* fluorescent reporters of signaling systems engaged by hunger would greatly enhance our understanding of how downstream circuitry is modulated in different internal states. This basic understanding of how hunger is included in behavioral computations will lay the foundation for future therapeutic and diagnostics development by generating novel targets. We hope that the predicted circuit mechanisms discussed here serve as an entry point to the study of hunger-driven changes in behavior.

Conclusions

Based on the numerous phenotypes presented here, we argue hunger must be recognized as an internal state able to alter a broad range of behaviors (Fig. 6). From its ability to encourage exploration and acquisition, to its ability to depress competing motivations from pain to social behaviors, it is clear hunger has a wide reach across circuits directing behavior. We are only just beginning to understand how this interoceptive state, so familiar to every animal, works within the brain to shift motivation. As we complete maps of the neural circuits that direct animal behavior, we propose that these circuit maps will serve as the foundation for studies of the state-dependent neuromodulation that

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results in the adaptive prioritization of behavior [100]. Understanding the modulatory mechanisms that direct neural traffic based on internal priorities will allow us to understand how and why specific circuits are selected, and thus, why certain behaviors are expressed. Furthering our knowledge of these mechanisms in the context of hunger will be integral to the production of novel therapeutics for disorders that result in maladaptive responses to energy deficit or surplus, and will also prove useful to modulate non-feeding behaviors in other disorders as well.

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Conflict of interest

The authors declare no conflict of interest.

Author contributions

NKS drafted the manuscript. NKS and BAG edited and revised the final version.

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